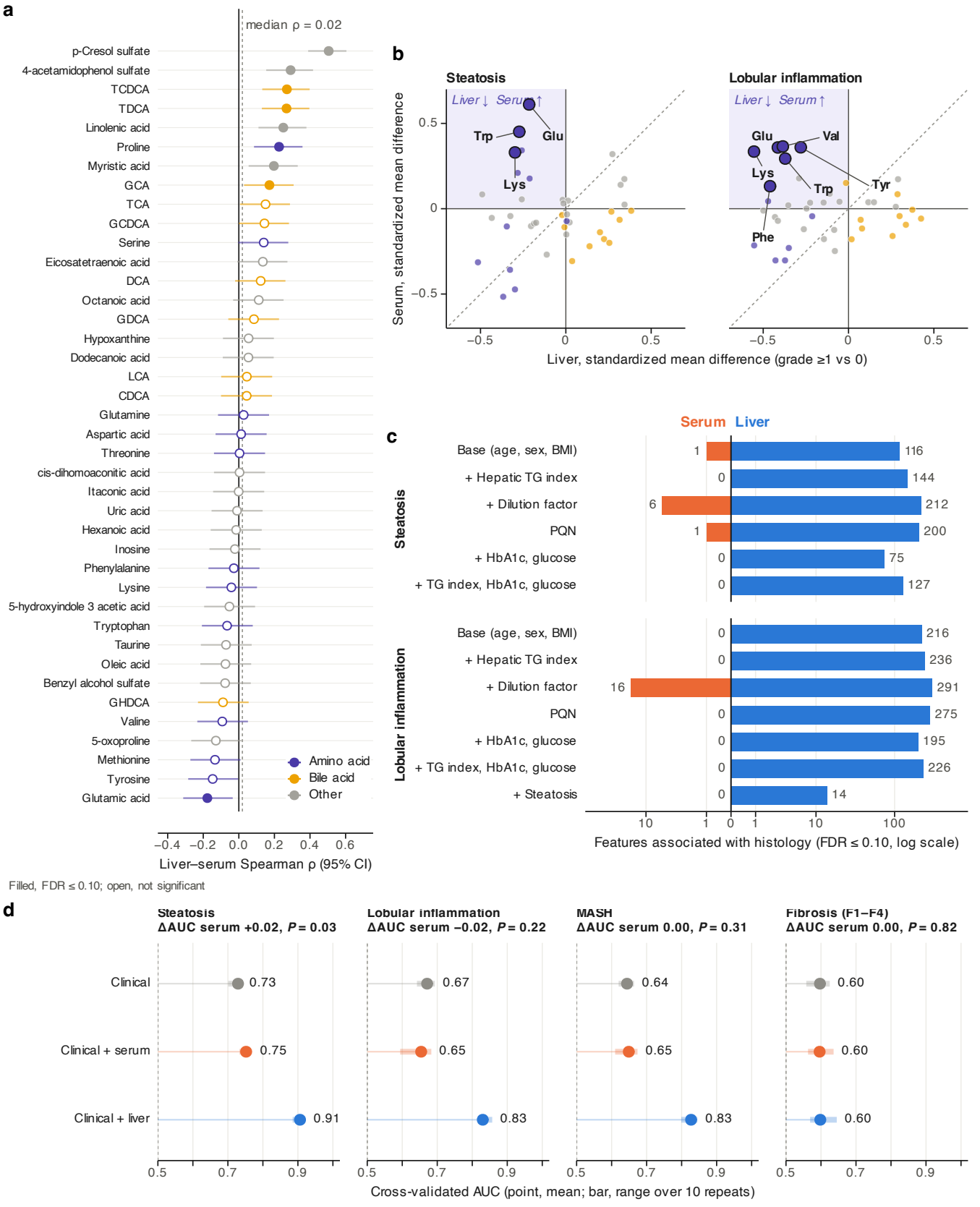
a
median ρ = 0.02
p-Cresol sulfate
4-acetamidophenol sulfate
TCDCA
TDCA
Linolenic acid
Proline
Myristic acid
GCA
TCA
GCDCA
Serine
Eicosatetraenoic acid
DCA
Octanoic acid
GDCA
Hypoxanthine
Dodecanoic acid
LCA
CDCA
Glutamine
Aspartic acid
Threonine
cis-dihomoaconitic acid
Itaconic acid
Uric acid
Hexanoic acid
Inosine
Phenylalanine
Lysine
5-hydroxyindole 3 acetic acid
Tryptophan
Taurine
Oleic acid
Benzyl alcohol sulfate
GHDCA
Valine
5-oxoproline
Methionine
Tyrosine
Glutamic acid
Amino acid
Bile acid
Other
−0.4 −0.2 0.0 0.2 0.4 0.6
Liver–serum Spearman ρ (95% CI)
Filled, FDR ≤ 0.10; open, not significant
b
Steatosis
Liver ↓ Serum ↑
Trp
Glu
Lys
Lobular inflammation
Liver ↓ Serum ↑
Glu
Val
Lys
Trp
Tyr
Phe
Serum, standardized mean difference
0.5
0
−0.5
−0.5
0
0.5
Liver, standardized mean difference (grade ≥1 vs 0)
c
Serum
Liver
Steatosis
Base (age, sex, BMI) 1 116
+ Hepatic TG index 0 144
+ Dilution factor 6 212
PQN 1 200
+ HbA1c, glucose 0 75
+ TG index, HbA1c, glucose 0 127
Lobular inflammation
Base (age, sex, BMI) 0 216
+ Hepatic TG index 0 236
+ Dilution factor 16 291
PQN 0 275
+ HbA1c, glucose 0 195
+ TG index, HbA1c, glucose 0 226
+ Steatosis 0 14
10 1 0 1 10 100
Features associated with histology (FDR ≤ 0.10, log scale)
d
Steatosis
ΔAUC serum +0.02, P = 0.03
Lobular inflammation
ΔAUC serum −0.02, P = 0.22
MASH
ΔAUC serum 0.00, P = 0.31
Fibrosis (F1–F4)
ΔAUC serum 0.00, P = 0.82
Clinical
Clinical + serum
Clinical + liver
0.73
0.75
0.91
0.67
0.65
0.83
0.64
0.65
0.83
0.60
0.60
0.60
0.5 0.7 0.9
Cross-validated AUC (point, mean; bar, range over 10 repeats)